ActiveVisionBenchmark: Stimulus Construction for Visual-Flow Orientation Experiments

Laurent Perrinet

Institut de Neurosciences de la Timone, CNRS & Aix-Marseille Université, France

June 29, 2026

Introduction

ActiveVisionBenchmark is a compact Python library whose primary purpose is to construct visual stimuli, with a specific emphasis on oriented visual-flow patterns.

The reference stimulus design is documented in: Orienting yourself in the visual flow.

Rather than positioning the project mainly as a benchmark suite, the library is now centered on reproducible stimulus creation for experimental pipelines in behavioral, neurophysiological, and computational vision studies.

Stimulus definition

The implemented stimulus is based on three principles:

1. A visual-flow field defines local motion direction around a configurable focus of expansion.
2. Local oriented elements are generated from that flow direction, with a configurable angular offset.
3. The final stimulus is composited on a controlled-noise background with explicit random seeds for reproducibility.

This formulation makes it straightforward to manipulate interpretable variables (density, orientation offset, flow origin, contrast) while preserving a compact single-file implementation.

Library scope

The library includes:

- low-level generators (noise and Gabor primitives),
- a dedicated visual-flow stimulus constructor,

- optional benchmarking utilities retained for downstream model-comparison workflows.

Conclusion

ActiveVisionBenchmark should be used first as a stimulus-construction library. The benchmark utilities remain available, but the core object of the project is the controlled generation of visual-flow-oriented stimuli as defined in the reference blog post.

References